Introduction to Computer and Computer Science

Homework Assignment #11

Due: 12/18 0:00 am

Balance Sheet

In order to manage your asset wisely, you might want to have a data frame, where you can record any information about your money, such as incomes, expenses, and cash flow. Such data frame is called the "balance sheet." In this homework, a balance sheet called "HW#11_data.csv" is provided and it records the daily profits of 10 merchants ("A," "B," "C," …, and "J") over the last 20 days. We are going to use "pandas" to deal with this balance sheet and then return some outcomes.

Enjoy it!

Problem #1: Read a .csv file

Please use the method "**DataFrame.read_csv**" in **pandas** to import "**HW#11_data.csv**" into your program and assign it as a DataFrame object. You should acquire this result when you print it:

```
   Day #1  Day #2  Day #3  Day #4  ...   Day #17  Day #18  Day #19  Day #20
A   -78.0   -22.0   -77.0   112.0  ...    -169.0     -9.0    115.0   -120.0
B    38.0     9.0   -18.0   -84.0  ...     -19.0     71.0    212.0     42.0
C    23.0   -92.0   -83.0   111.0  ...     -89.0    -19.0    -16.0     23.0
D   -65.0    72.0     3.0    52.0  ...    -142.0    -40.0    -74.0     41.0
E   134.0   -12.0    39.0    22.0  ...      15.0    -71.0    149.0    159.0
F   -43.0    45.0   -94.0    16.0  ...     -38.0    -29.0    -69.0    100.0
G   101.0    11.0  -142.0    31.0  ...      45.0    133.0    -56.0   -220.0
H  -104.0    42.0   -28.0    17.0  ...     -95.0     97.0     12.0    -69.0
I   -66.0    55.0    11.0   -17.0  ...     189.0    160.0    -84.0     32.0
J   168.0   -82.0   112.0   -30.0  ...    -101.0   -127.0   -115.0    -17.0

[10 rows x 20 columns]
```

Hint: Please be aware! The index of the DataFrame you have assigned should be the first column of "**HW#11_data.csv**."

Problem #2: Data analysis

Please write a function called "**analysis**" to calculate the total profits of each merchants over the last 20 days, and add an additional column named "Profit" to record the corresponding results. Meanwhile, since every one wants to be rich, let's assume that the target profit of each merchants is "**target_profit**." Please also calculate how much money they should earn per day during the next 10 days so that they can meet the target profit. The parameter of your function should be,

$$\textbf{analysis}(\text{df},\ \text{target\_profit} = 100),$$

where "df" is the input data (should be a DataFrame object as shown in problem #1) of this function and target_profit represents the corresponding target profit as previous mentioned (let's set 100 by default).

This function should return you a new DataFrame object, which consists of the original data and two additional columns named "Profit" and "Required" as shown below:

Case 1

target_profit = 100

```
   Day #1  Day #2  Day #3  Day #4    ...     Day #19  Day #20  Profit  Required
A   -78.0   -22.0   -77.0   112.0    ...       115.0   -120.0   143.0       0.0
B    38.0     9.0   -18.0   -84.0    ...       212.0     42.0   589.0       0.0
C    23.0   -92.0   -83.0   111.0    ...       -16.0     23.0   185.0       0.0
D   -65.0    72.0     3.0    52.0    ...       -74.0     41.0   217.0       0.0
E   134.0   -12.0    39.0    22.0    ...       149.0    159.0   -30.0      13.0
F   -43.0    45.0   -94.0    16.0    ...       -69.0    100.0  -462.0      57.0
G   101.0    11.0  -142.0    31.0    ...       -56.0   -220.0  -347.0      45.0
H  -104.0    42.0   -28.0    17.0    ...        12.0    -69.0  -249.0      35.0
I   -66.0    55.0    11.0   -17.0    ...       -84.0     32.0   -10.0      11.0
J   168.0   -82.0   112.0   -30.0    ...      -115.0    -17.0  -285.0      39.0

[10 rows x 22 columns]
```

Case 2

target_profit = 500

```
   Day #1  Day #2  Day #3  Day #4    ...     Day #19  Day #20  Profit  Required
A   -78.0   -22.0   -77.0   112.0    ...       115.0   -120.0   143.0      36.0
B    38.0     9.0   -18.0   -84.0    ...       212.0     42.0   589.0       0.0
C    23.0   -92.0   -83.0   111.0    ...       -16.0     23.0   185.0      32.0
D   -65.0    72.0     3.0    52.0    ...       -74.0     41.0   217.0      29.0
E   134.0   -12.0    39.0    22.0    ...       149.0    159.0   -30.0      53.0
F   -43.0    45.0   -94.0    16.0    ...       -69.0    100.0  -462.0      97.0
G   101.0    11.0  -142.0    31.0    ...       -56.0   -220.0  -347.0      85.0
H  -104.0    42.0   -28.0    17.0    ...        12.0    -69.0  -249.0      75.0
I   -66.0    55.0    11.0   -17.0    ...       -84.0     32.0   -10.0      51.0
J   168.0   -82.0   112.0   -30.0    ...      -115.0    -17.0  -285.0      79.0

[10 rows x 22 columns]
```

Hint: Only integer is considered.

Good luck and enjoy it!

Please be aware, the teaching assistants will use "import" command to import your program and execute some testing code to examine your program. **The name of your functions, class object, including the name of corresponding methods, should be exactly as same as the name we shown here**. Otherwise it will pop out an error and terminate the testing procedure. In that case, your homework will be considered to be failed. **Please double check your program carefully**.

Hand in procedure:

As we had mentioned in the lecture, you should list all of your collaborators in your programs. Here is the template:

```
# ==========================================================================
#
# Homework assignment #0
# Student: Your name, 你的名字
# Student ID: Your student ID, 你的學號
# Collaborators: John Doe, Jane Doe, ...
#
# ==========================================================================
# Your code goes here...

```

Please save your code as a ".py" file, where the file name should follow this format:

108A_hw#?_YourID.py

For example,

108A_hw#1_0180506.py

Please be aware. **We are not going to accept any homework file with wrong file name or without signature**. Please double check the content of your file.

Once you have accomplished your works, you can upload your homework to the "New e3" system. There will be a section for uploading your homework.